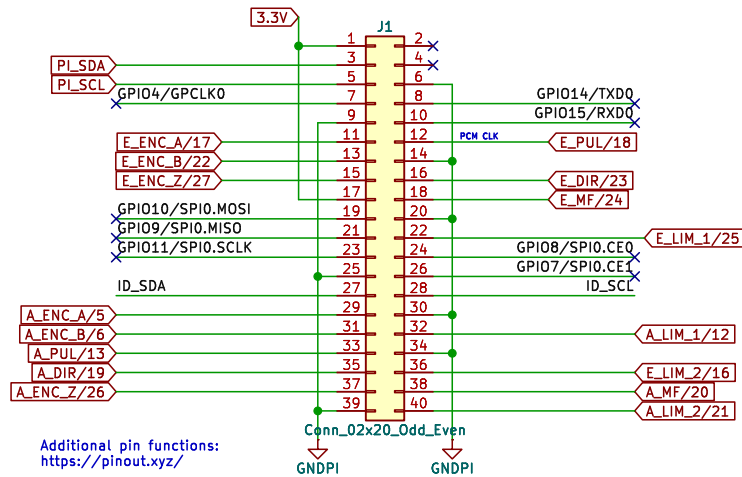


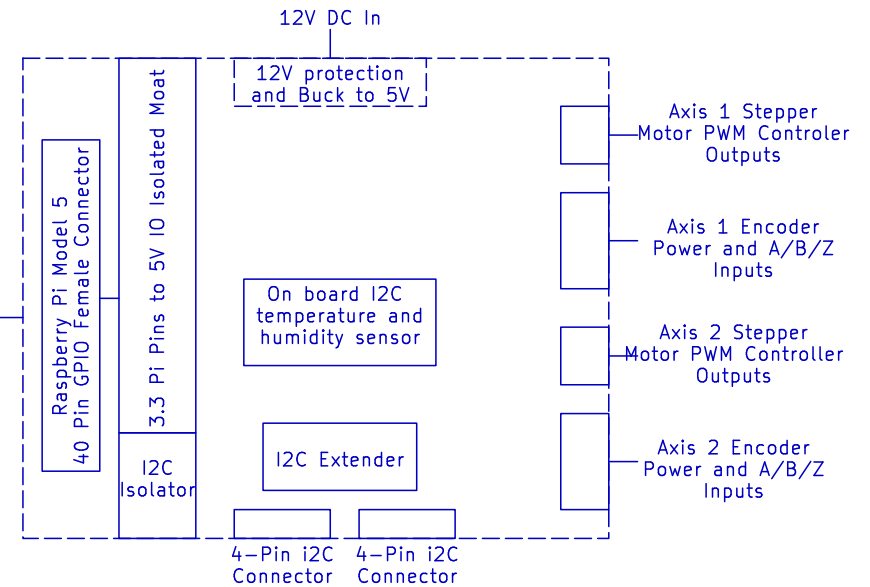
Raspberry Pi Hat 2 Axis Stepper Motor Controller For Satellite Dish

The Pi is powered from the USB-C connector on the Pi

Raspberry Pi GPIO Pin Connections



Raspberry Pi
With PID loop
Controlling Motor
PWM and reading
encoder inputs

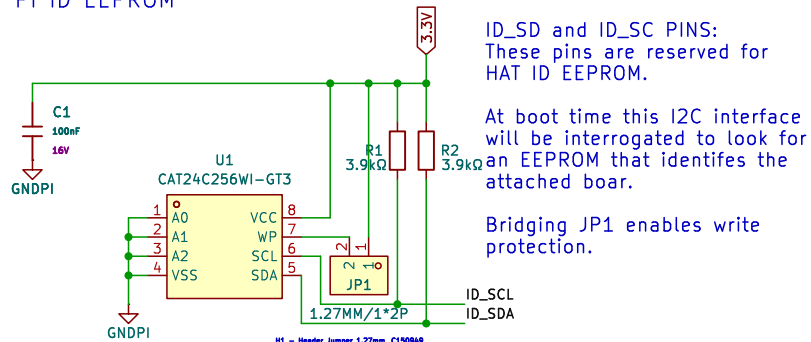


This PCB controls the Azimuth and Elevation axes of a satellite dish. This is a DIY project for now. The target application is amateur radio astronomy.

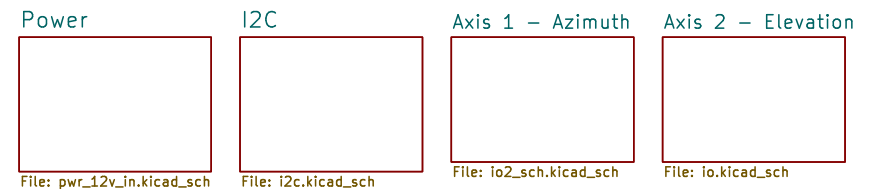
The initial dish for testing is a 3-meter satellite dish that needs precise pointing of under 1 degree accuracy for high frequencies from 1.2GHz to 8.4GHz.

The PCB outputs PWM to control a stepper motor. Feedback is from an encoder on the axis. The Raspberry Pi uses a PID loop to adjust the speed and position of the motor.

Pi ID EEPROM



I2C sensors placed at each axis monitor temperature and humidity at the motors and gears.



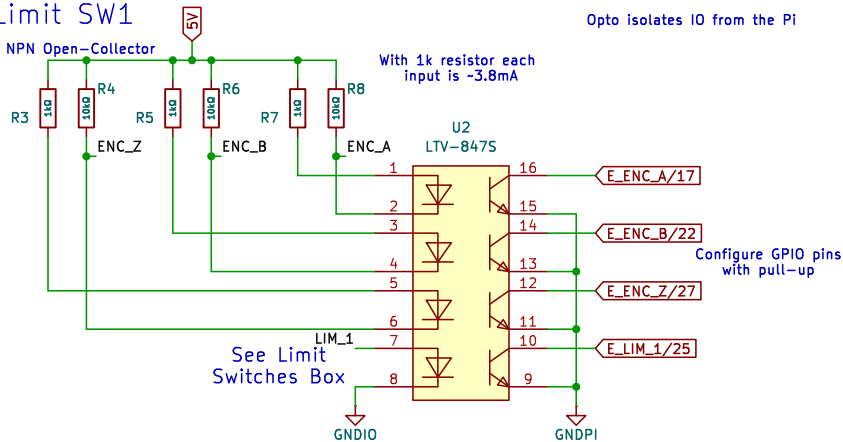
Elevation Axis Circuits

Connects to a motor controller like the DM556Y

Reads elevation axis controller type E6B2-CWZ6C 2000 PPR, 5V

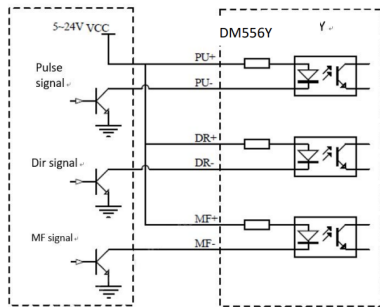
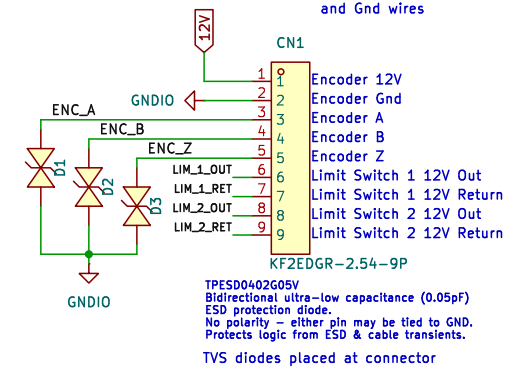
El Encoder Input Plus Limit SW1

Encoder is NPN Open-Collector

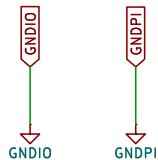


Encoder Input Off: No current flows, pulled high, no LED current
Encoder Input ON: Current flows from 5V thru LED to encoder

Terminal Block – Encoder and Limit Switches

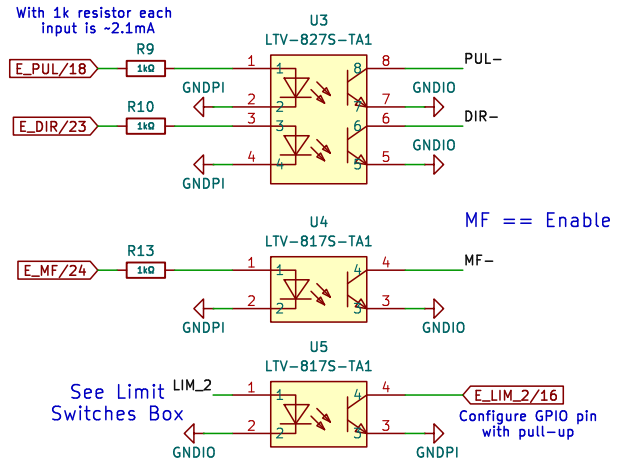


How to wire from the DM556Y manual. When Vcc is 5V there is no series resistor.



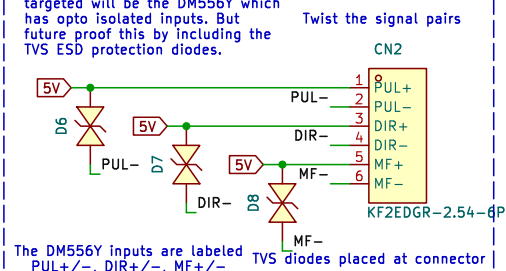
El Motor Pulse/Dir/Enable Plus Limit Sw2

Opto isolates IO from the Pi

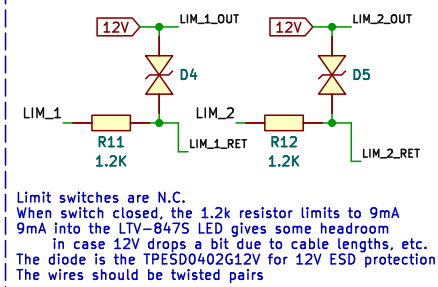


Motor Controller Signals

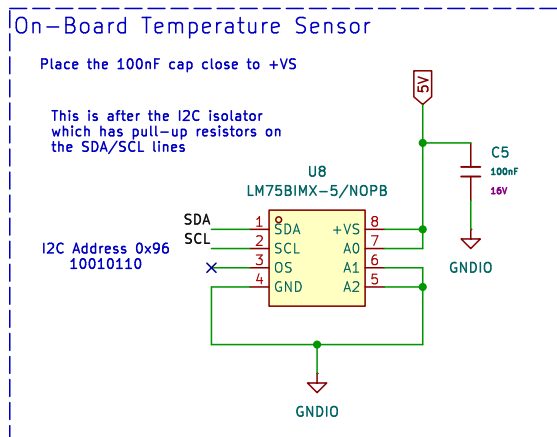
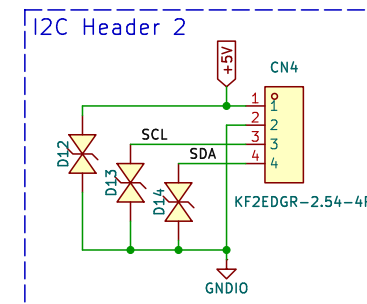
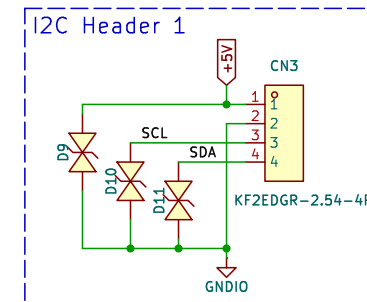
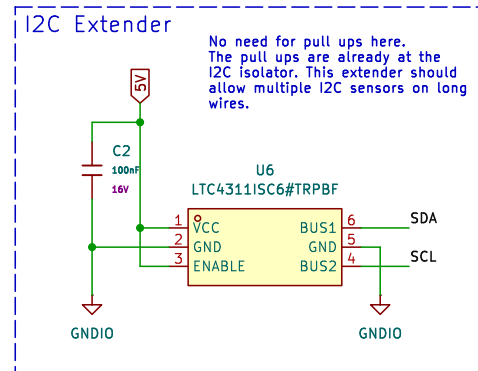
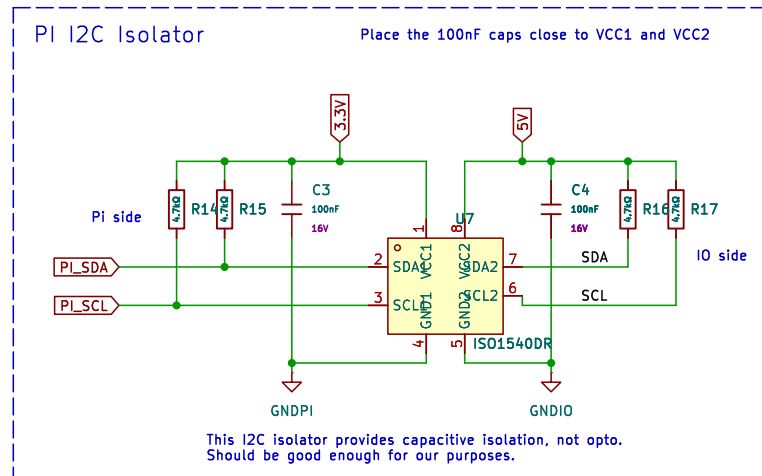
The motor controller first targeted will be the DM556Y which has opto isolated inputs. But future proof this by including the TVS ESD protection diodes.



Limit Switches



I2C Circuits



TPESD0402G05V
 Bidirectional ultra-low capacitance (0.05pF)
 ESD protection diodes on input connectors.
 No polarity – either pin may be tied to GND.
 Protects from ESD & cable transients.

Pi 3.3V
 I2C

3.3V to
 5V I2C
 isolation

On Board
 Temp and
 Humid
 I2C

I2C
 Extender
 LTC4311

On Axis
 I2C
 Sensors

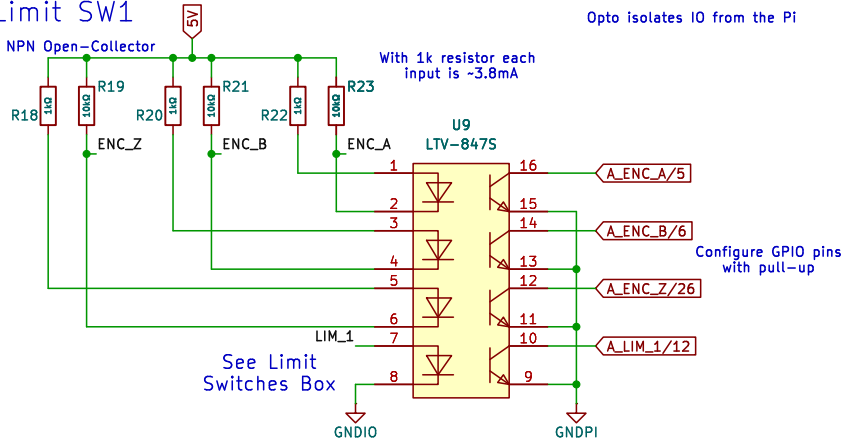
Azimuth Axis Circuits

Connects to a motor controller like the DM556Y

Reads azimuth axis controller type E6B2-CWZ6C 2000 PPR, 5V

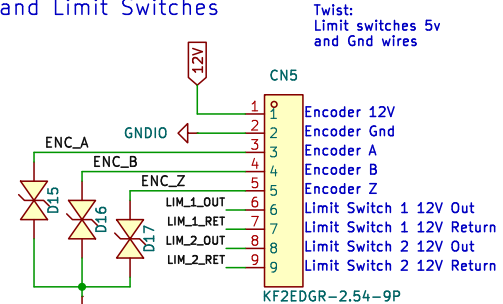
Az Encoder Input Plus Limit SW1

Encoder is NPN Open-Collector

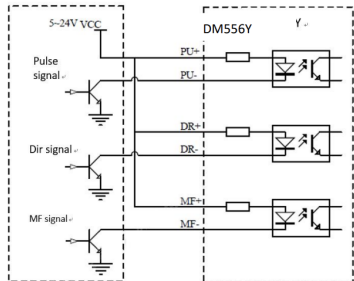


Encoder Input Off: No current flows, pulled high, no LED current
Encoder Input ON: Current flows from 5V thru LED to encoder

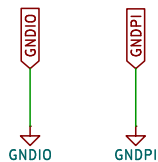
Terminal Block – Encoder and Limit Switches



TPESD0402G05V
Bidirectional ultra-low capacitance (0.05pF)
ESD protection diode.
No polarity – either pin may be tied to GND.
Protects logic from ESD & cable transients.
TVS diodes placed at connector

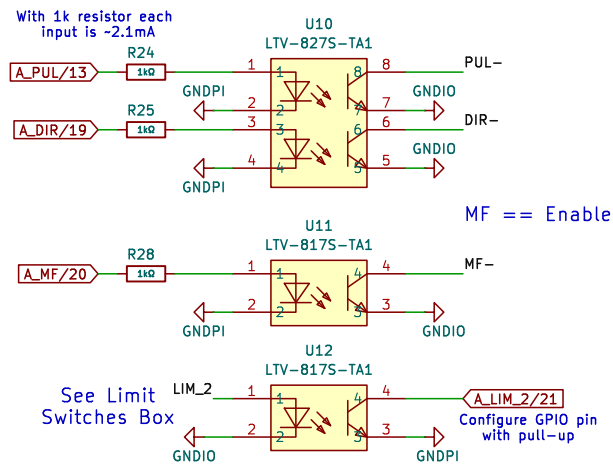


How to wire from the DM556Y manual. When Vcc is 5V there is no series resistor.



Az Motor Pulse/Dir/Enable Plus Limit Sw2

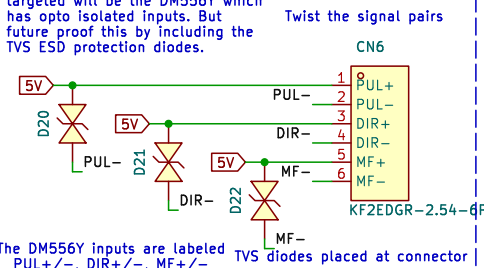
Opto isolates IO from the PI



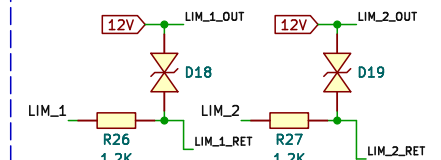
Split into multiple ICs to be able to geometrically fit into the PCB

Motor Controller Signals

The motor controller first targeted will be the DM556Y which has opto isolated inputs. But future proof this by including the TVS ESD protection diodes.

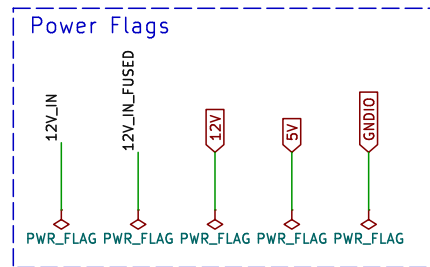


Limit Switches

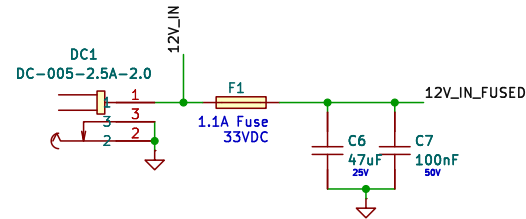


Limit switches are N.C.
When switch closed, the 1.2k resistor limits to 9mA into the LTV-8475 LED gives some headroom in case 12V drops a bit due to cable lengths, etc.
The diode is the TPESD0402G12V for 12V ESD protection
The wires should be twisted pairs

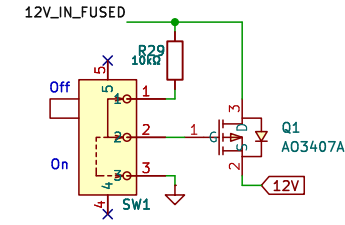
Power



12V DC input fused to 1.1A max



Reverse Voltage Protection
P-Channel MOSFET

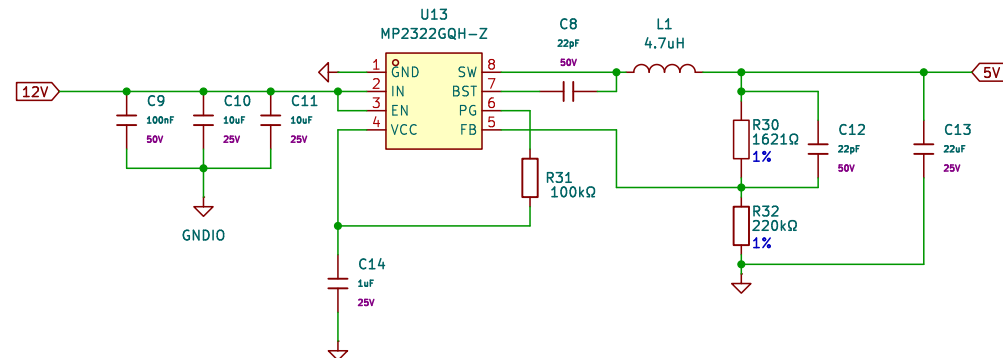


Switch: In Off pos. Gate is pulled up to 12V. Mosfet is off
In On pos. Gate is pulled to Gnd and the Mosfet is on

Notes:

Circuit to convert 12V DC to 5V DC
12V is reverse polarity and over voltage protected
MP2322 used to convert 12V to 5V DC
12V DC out will draw max 200mA, powering each encoder
5V DC out will draw max 300mA and power the board
Buck modeled after John Teel's YouTube video:
<https://www.youtube.com/watch?v=rLHW4gU6idU>
and referencing the MP2322 datasheet

12V to 5V Buck



From MP2322 Datasheet:

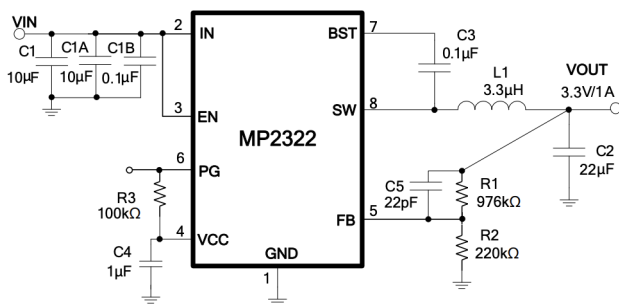


Figure 8: $V_{IN} = 12V$, $V_{OUT} = 3.3V$, $I_{OUT} = 1A$

Circuit the same except this circuit output is 5V, R1 and L1 are modified for 5V output →

voltages				
V_{OUT} (V)	R1 (kΩ)	R2 (kΩ)	L (μH)	C5 (pF)
1.0	147	220	2.2	22
1.2	220	220	2.2	22
1.8	440	220	3.3	22
2.5	697	220	3.3	22
3.3	976	220	3.3	22
5	1613	220	4.7	22